

Mesh Networking

Introduction

- Wireless Mesh Networks (WMNs) are built by interconnecting internet egress points with end-user devices can act as terminals and as routers.
- Mesh nodes are small radio transmitters that function in the same way as a wireless router.
- Nodes use the common WiFi standards known as 802.11a, b and g to communicate wirelessly with users, and, more importantly, with each other.

Wireless mesh networks can easily, effectively and wirelessly connect entire cities using inexpensive, existing technology



A node on a streetlight.

In a wireless mesh network, the network connection is spread out among dozens or even hundreds of wireless mesh nodes that "talk" to each other to share the network connection across a large area.

The Internet: The Original Mesh Network

- The Internet is the world's largest mesh network. Information travels across the Internet by being bounced automatically from one router to the next until it reaches its destination.
- The Internet is often depicted as a "cloud" or "web" of connectivity because of the billions of potential paths across which data can travel.

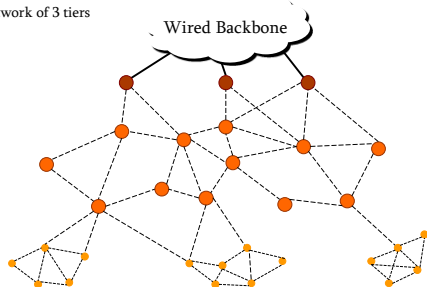
Applications

- Developing Countries
- Isolated Locations
- Healthcare
- Temporary Venues
- Future Applications
The U.S. military, which helped develop wireless mesh technology, foresees a day when thousands of microchip-size mesh nodes can be dropped onto a battlefield to set up instant scouting and surveillance networks.
Information can be routed to both ground troops and headquarter personnel.

- Carmakers and telecom companies are working to develop Intelligent Transport Systems (ITS) powered by street and highway-based wireless mesh networks.
- Using an automated network of surveillance cameras and in-car sensors, public safety officials can tightly monitor traffic accidents and dangerous road conditions.

How does this change look like?

A wireless mesh network of 3 tiers



Wireless mesh networks advantages include:

- Using fewer wires means it costs less to set up a network, particularly for large areas of coverage.
 - The more nodes you install, the bigger and faster your wireless network becomes.
 - They rely on the same WiFi standards (802.11a, b and g) already in place for most wireless networks.
 - They are useful for Non-Line-of-Sight (NLoS) network configurations.
- For example, if the blocks the signal from a wireless access point. If there are dozens or hundreds of other nodes around, the mesh network will adjust to find a clear signal.

- Mesh networks are "self configuring;" the network automatically incorporates a new node into the existing structure without needing any adjustments by a network administrator.
- Mesh networks are "self healing," since the network automatically finds the fastest and most reliable paths to send data, even if nodes are blocked or lose their signal.
- Wireless mesh configurations allow local networks to run faster, because Local packets don't have to travel back to a central server.
- Wireless mesh nodes are easy to install and uninstall, making the network extremely adaptable and expandable as more or less coverage is needed.

Standards & Committees

IEEE standards groups actively working to define specs. for wireless mesh networking techniques

Special groups established to define the requirements for mesh networking in WPANs, WLANs, and WMANs.

The following standards amendments are expected:

802.15.5	(bluetooth - WPAN)
802.11s	(wi-fi - WLAN)
802.16a	(wi-max - WMAN)

Also 802.20 (wireless mobile broadband access -WMBA) is to support the Mesh Networking

802.11s

* Scope of the Project: An IEEE 802.11 **Extended Service Set (ESS) Mesh** is a collection of APs interconnected with wireless links that enable automatic topology learning and dynamic path configuration.

802.16a

802.16 is a WMAN connection standard. (data rate up to 120Mbps @ 30miles).

802.16 approved in 2001 (10-66GHz operation – TDMA, TDD&FDD)

802.16a approved in Jan. 2003 (2-11 GHz operation added -OFDM)

Stations may have direct links to each other - control can be distributed.

WiMax forum formed later in 2003 (just like the 802.11 Wi-Fi forum) to promote IEEE standards for interoperability.

Major research activity by Microsoft...

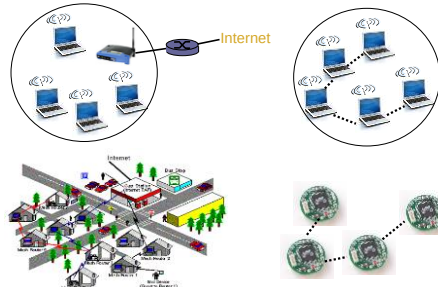
One of the most significant research efforts in the WMN field is conducted by the networking research group of Microsoft

In June Microsoft organized the "Mesh Networking Summit 2004"

Gathering most of the major players from the industry and academia community this was the first focused event on WMNs



Wireless networks



■ Wireless LAN, ad-hoc network, wireless mesh, sensor network